What is Learning?

San José State University

Agenda

1. (Informal) definition of “learning”
2. Fictitious play - Simple learning
3. Intro to Reinforcement Learning

1. (Informal) definition of “learning”

Let’s talk about “learning”

Rock Paper Scissors

- Past ten times
  - R P R R R P S R R R
  - Which hand do you choose for the 11th trial?
- Let’s play another eleven times now
  -

So how did you learn?

- Discussion

So how did you learn?

- Discussion
- Offline vs Online
- Static vs Dynamic

Characteristics of “Learning”

- “Learning” setting involves some temporal nature of the domain and the regularity across time
- An agent’s next choice depends on the history of play
- The agent could guess that the frequency of actions played by his opponent in the past and play a best response to that mixed strategy.

More Knowledge, Better Performance?

- One consequence is that learning in the sense of “accumulated knowledge" is not always beneficial.
- In the abstract, accumulating knowledge never hurts, since one can always ignore what has been learned.
- But when one pre-commits to a particular strategy for acting on accumulated knowledge, sometimes less is more.

Optimal Learning

- No learning procedure is optimal against all possible opponent behaviors.
- Indeed, in the broad sense in which we use the term, a “learning strategy” is simply a strategy in a game that happens to have a component that is naturally viewed as adaptive.
- How can we evaluate “learning strategies”?

How do we evaluate learning?

- Ask whether a learning strategy achieves payoffs that are “high enough”
  - **Safety:** A learning rule is safe if it guarantees the agent at least its maxmin payoff, or “security value.”

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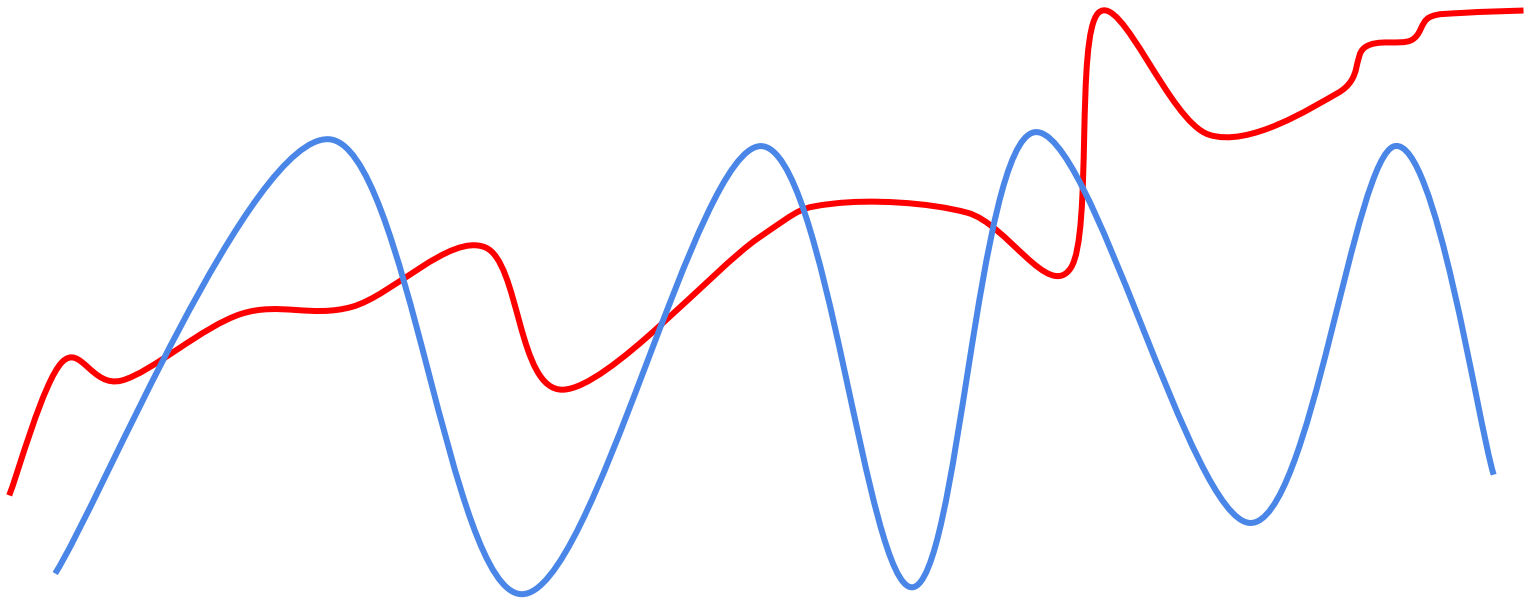
  - **Safety:** A learning rule is safe if it guarantees the agent at least its maxmin payoff, or “security value.”
  - **Rationality:** A learning rule is rational if whenever the opponent settles on a stationary strategy, the agent settles on a best response to that strategy.
  - **No-regret:** a strategy that guarantees the player that in hindsight no single action when played throughout the game would have performed significantly better

Regret Example



Milestone Questions

If you trained a stock trading agent to seek the safety measure, which strategy would it choose?

Stock A: The estimated value in 10 years ranges from $10 to $500
Stock B: The estimated value in 10 years ranges from $100 to $300
Stock C: The estimated value in 10 years is $180
Stock D: The estimated value in 10 years ranges from $200 to $400

2. Fictitious Play

Fictitious Play

- Simple learning method, which involves
  - Belief about the opponent
  - Observation of the actual plays
- Usually viewed as a model of learning

```
Initialize beliefs about the opponent's strategy
repeat
    Play a best response to the assessed strategy of the opponent
    Observe the opponent's actual play and update beliefs accordingly
```

Belief

- An agent believes that his opponent is playing the mixed strategy given by the empirical distribution of the opponent's previous actions.
- The agent assesses the probability of a in the opponent's mixed strategy as:

$$P(a) = \frac{w(a)}{\sum_{a' \in A} w(a')}.$$

- A: All actions in the past
- w(): # of times the opponent having taken the action a

Belief: An Example

- Rock Paper Scissors
- We observed the opponent played
  - R, R, P, S, S, R, P, R, S, S

Belief: An Example

- Rock Paper Scissors
- We observed the opponent played
  - R, R, P, S, S, R, P, R, S, S
  - R: 4
  - P: 2
  - S: 4
- The (learned) probability distribution
  - (R, P, S) = (0.4, 0.2, 0.4)
- The optimal action for us for the next round
  - P (against R), or R (against S)
- Note: The optimal changes with more observations

Initial Belief

- Initial belief (action counts that were observed before the start of the game) can have a radical impact on the learning process
  - The prior beliefs cannot be (0, . . . , 0) since this does not define a meaningful strategy
- As you can see, each player ends up alternating back and forth between playing heads and tails.
  - In fact, as the number of rounds tends to infinity, the empirical distribution of the play of each player will converge to (0.5, 0.5)

		Player 1: Heads	Player 1: Tails
Player 2	Heads	$1, -1$	$-1, 1$
	Tails	$-1, 1$	$1, -1$

Figure 7.4: Matching Pennies game.

Round	1's action	2's action	1's beliefs	2's beliefs
0			(1.5,2)	(2,1.5)
1	T	T	(1.5,3)	(2,2.5)
2	T	H	(2.5,3)	(2,3.5)
3	T	H	(3.5,3)	(2,4.5)
4	H	H	(4.5,3)	(3,4.5)
5	H	H	(5.5,3)	(4,4.5)
6	H	H	(6.5,3)	(5,4.5)
7	H	T	(6.5,4)	(6,4.5)
⋮	⋮	⋮	⋮	⋮

https://en.wikipedia.org/wiki/Matching_pennies

Milestone Questions

- Complete the table below

	A	B
A	$0,0$	$1,1$
B	$1,1$	$0,0$

Figure 7.5: The Anti-Coordination game.

Round	1's action	2's action	1's beliefs	2's beliefs
0			(1,0.5)	(1,0.5)
1				
2				
3				
4				
⋮	⋮	⋮	⋮	⋮

Another Example

	A	B
A	$0,0$	$1,1$
B	$1,1$	$0,0$

Figure 7.5: The Anti-Coordination game.

Round	1's action	2's action	1's beliefs	2's beliefs
0			(1,0.5)	(1,0.5)
1	B	B	(1,1.5)	(1,1.5)
2	A	A	(2,1.5)	(2,1.5)
3	B	B	(2,2.5)	(2,2.5)
4	A	A	(3,2.5)	(3,2.5)
⋮	⋮	⋮	⋮	⋮

Convergence of Fictitious Play

- The empirical play of this game never converges to any fixed distribution.

	Rock	Paper	Scissors
Rock	0, 0	0, 1	1, 0
Paper	1, 0	0, 0	0, 1
Scissors	0, 1	1, 0	0, 0

Figure 7.6: Shapley's Almost-Rock-Paper-Scissors game

Round	1's action	2's action	1's beliefs	2's beliefs
0			(0,0,0.5)	(0,0.5,0)
1	Rock	Scissors	(0,0,1.5)	(1,0.5,0)
2	Rock	Paper	(0,1,1.5)	(2,0.5,0)
3	Rock	Paper	(0,2,1.5)	(3,0.5,0)
4	Scissors	Paper	(0,3,1.5)	(3,0.5,1)
5	Scissors	Paper	(0,1.5,0)	(1,0,0.5)
⋮	⋮	⋮	⋮	⋮

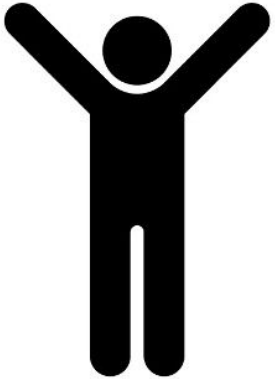
3. Intro to Reinforcement Learning

Learning by Interactions

- “We learn by interacting with our environment”
  - E.g. When an infant plays, waves its arms, or looks about, it has no explicit teacher, but it does have a direct sensorimotor connection to its environment
- Exercising this connection produces a wealth of information about
  - Cause and effect
  - The consequences of actions
  - What to do in order to achieve goals
- Throughout our lives, such interactions are undoubtedly a major source of knowledge about our environment and ourselves

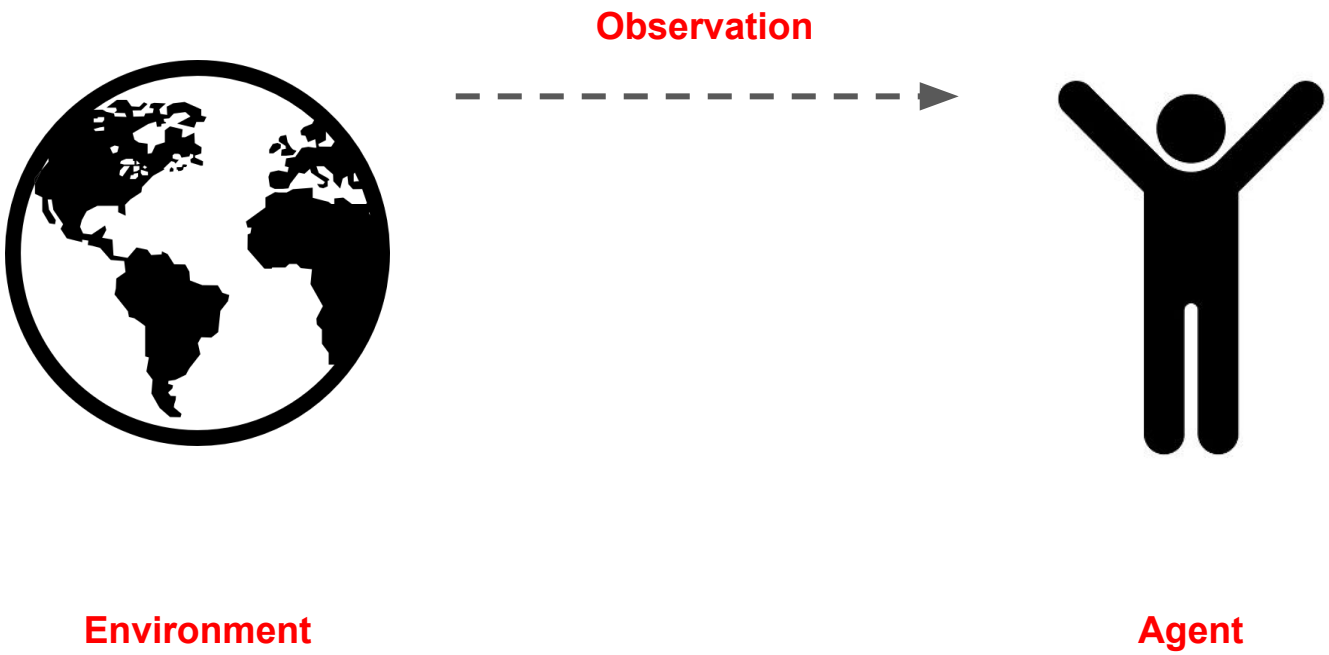
Learning to Take an Optimal Action

Environment

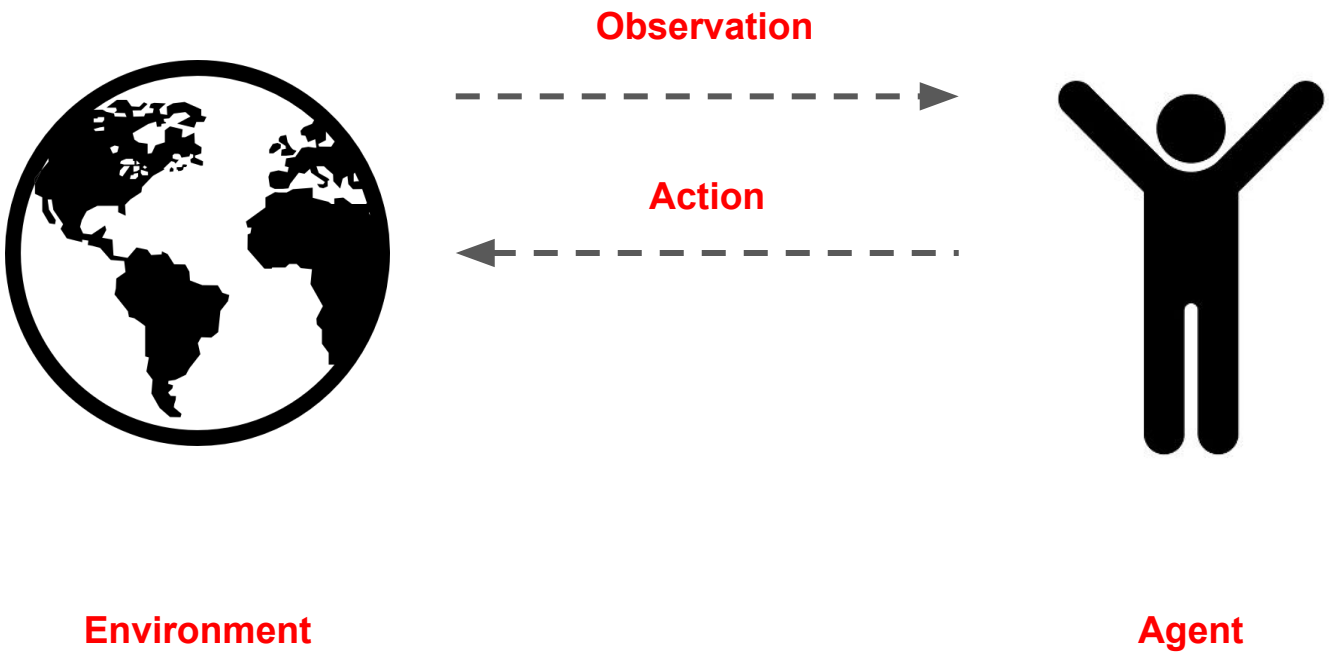


Agent

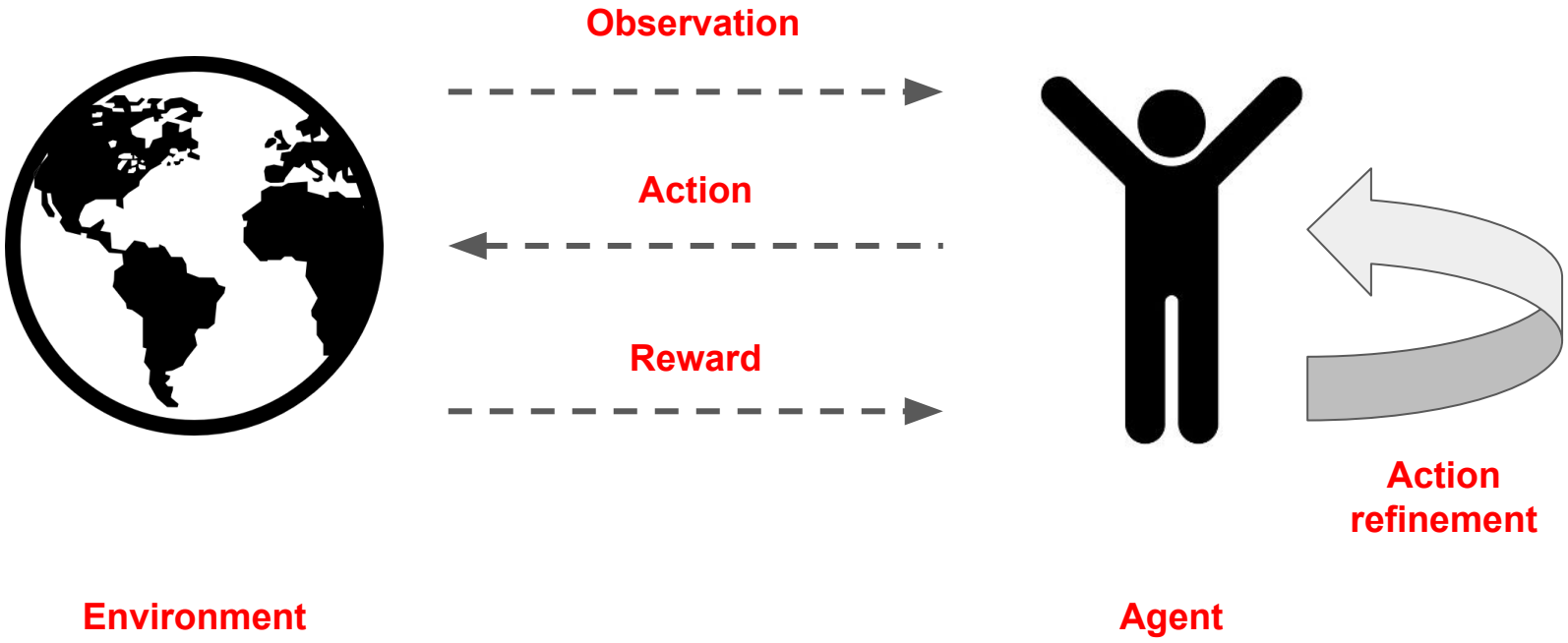
Learning to Take an Optimal Action



Learning to Take an Optimal Action



Learning to Take an Optimal Action



Reinforcement Learning

- A computational approach to learning from interaction
- Learning what to do—how to **map situations to actions**—so as to **maximize a numerical reward signal**
- Two important aspects
  - Trial-and-error search
  - Delayed reward: Actions may affect not only the immediate reward but also the next situation and, through that, all subsequent rewards (Can you think of any example?)

Trial-and-Error Search

- Exploration: You try out new actions to find better solutions
- Exploitation: You use a sequence of actions that performed well in the past runs

Delayed Rewards

- I woke up early last Sunday and finished some questions of an assignment due on Friday.
- I did not have a super relaxing Sunday.
- I had enough time to finish the rest of the assignment during weekdays and review all the answers too.
- I got a good grade.
- I also feel I am ready for coming exams.

Delayed Rewards

- I slept till 1PM last Sunday and enjoyed the relaxing time!!
- BUT I did not have time for an assignment due on Friday.
- I needed to finish the assignment under limited time during weekdays and did not have time to review all answers.
- I got a failing grade.
- I need to spend more time to study during the weekends for coming exams

An Example: Atari Breakout

Current Decision and Future Possibilities

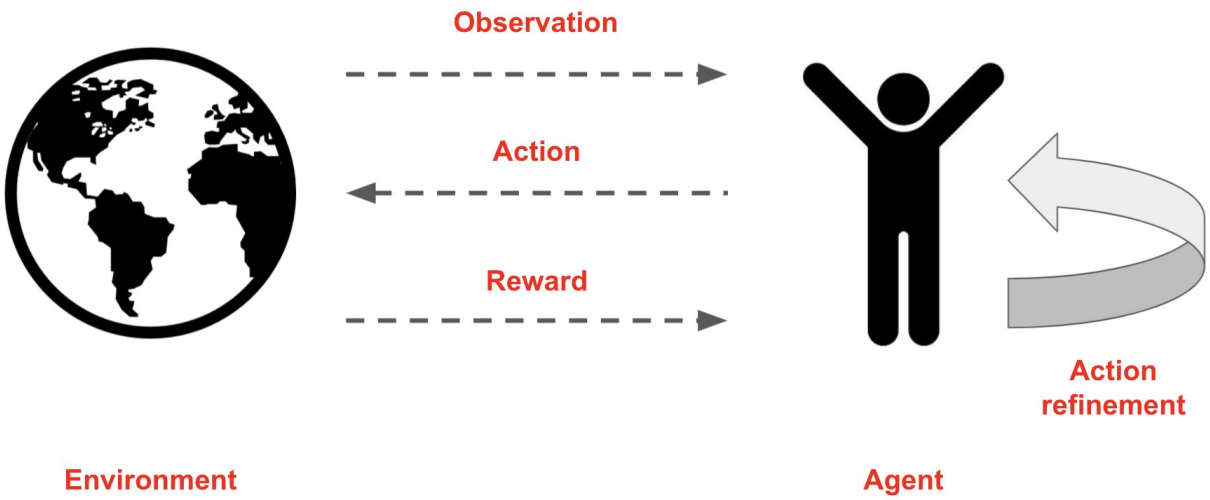
- All involve interaction between an active decision-making agent and its environment,
  - The agent seeking to achieve a goal
  - Uncertainty about its environment
- The agent's actions are permitted to affect the future state of the environment
  - E.g. The next chess position
  - E.g. The robot's next location & future charge level of its battery
  - I.e. The actions and opportunities available to the agent at later times are affected by the current action

Key Concepts

- State
- Action
- Reward (reward signal)
- Policy
- Model

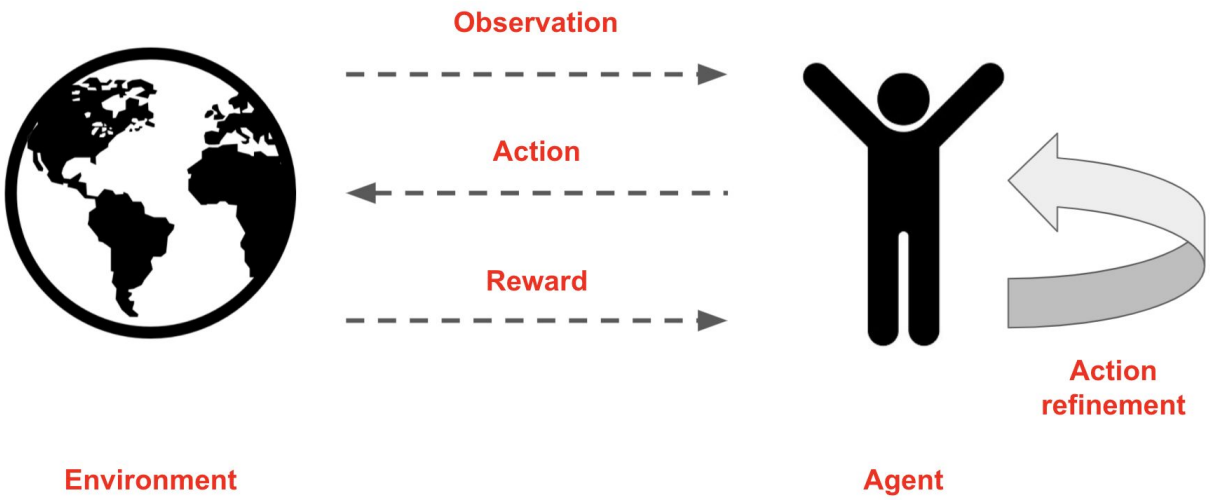
Key Concepts: State/Action

- An action is a method that allows an agent to interact and change its environment
- A state of an environment is a representation of the current world
- Note: A state is not always same as an observation



Key Concepts: Reward

- A reward is a numerical feedback signal
- It indicates how well the agent is doing at that round (immediate feedback)
- Reinforcement learning is based on “reward hypothesis”



Key Concepts: Reward Hypothesis

All goals (practical tasks to learn) can be described by the maximization of expected cumulative reward

E.g. Manage an investment portfolio

- A monetary reward (current balance)
- Investment distribution

E.g. Make a humanoid robot walk

- Positive reward for moving forward
- Negative reward for falling
- Joint movement planning

Delayed Rewards Revisited

- You may want to sacrifice immediate positive reward for a bigger future reward
  - Actions may have long term consequences (Future available actions may be limited by past actions)
  - Can you think of any example?

Delayed Rewards Revisited

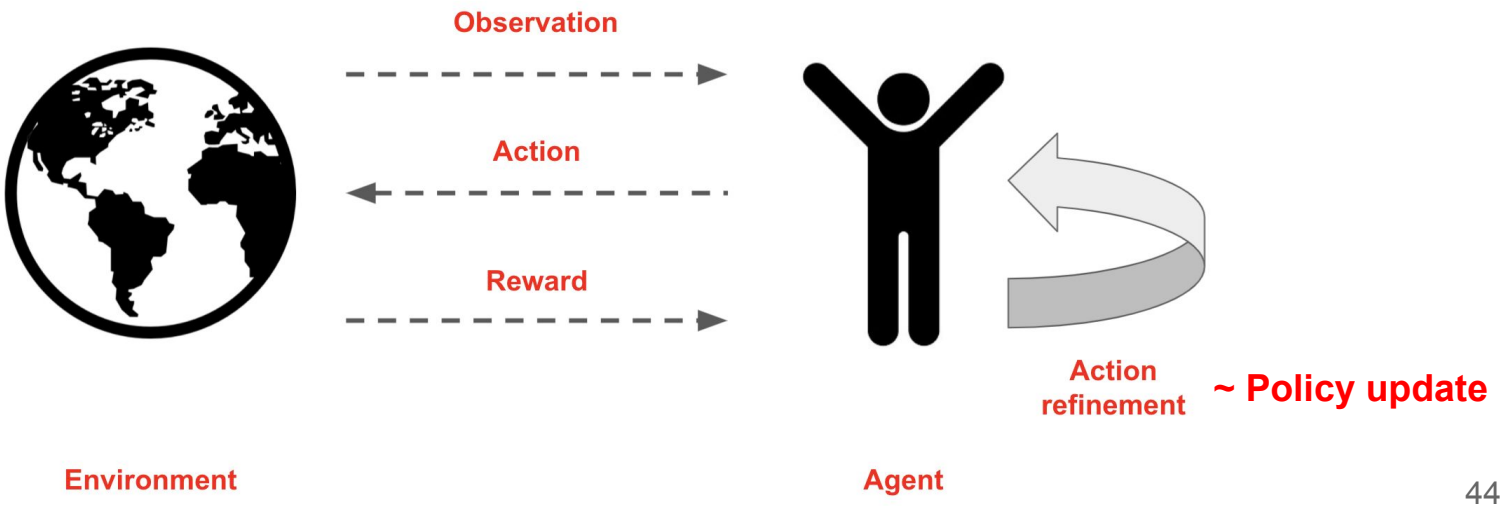
- You may want to sacrifice immediate positive reward for a bigger future reward
  - Actions may have long term consequences (Future available actions may be limited by past actions)
  - Can you think of any example?
    - Buying a cup of Starbucks almost everyday $5 x 300 days = $1,500
    - Saving and buy a new macbook

Key Concepts: Policy

- Mapping from states to actions
- I.e. decide an so-far best action for a given observation (state)
- Note: we are trying to learn an optimal policy through RL

Key Concepts: Model

- A theoretical entity that mimics the behavior of the environment
- More generally, a model allows inferences to be made about how the environment will behave given a state and an action



Key Concepts: Model

E.g. Rock paper scissors

- If you know an exact probability distribution over R/P/S (model),
- you can use the knowledge to devise an optimal policy

This approach is called ***planning***, which does NOT involve try-and-error searches

0.5 0.2 0.3

Key Concepts: Model

E.g. Rock paper scissors

- If you do NOT know a probability distribution over R/P/S,
- you need to figure out an optimal policy by observing the opponent

This approach is called ***model-free methods***, *which DO involve try-and-error searches*

? ? ?

Milestone Questions

- True or False: A reward is either 0 (bad) or 1 (good).

Milestone Questions

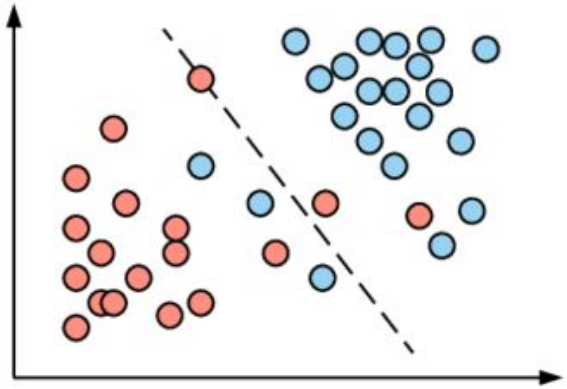
- Complete the following sentence about a policy of an agent
  - A policy is a function that maps (X) and (Y)

Milestone Questions

- Explain the difference between an action and a policy

Difference between RL and other ML

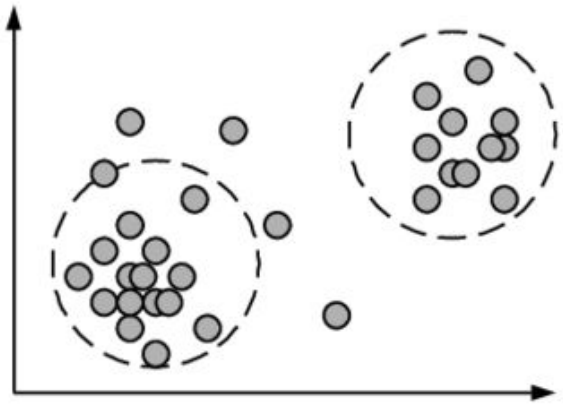
- Supervised learning: learning from a training set of labeled examples provided by a knowledgeable external supervisor.
  - Each example: a description of a situation together with a specification—the label—of the correct action
- RL problems: learning from interactions even in uncharted situations



Supervised learning

Difference between RL and other ML

- Unsupervised learning: learning of a structure hidden in collections of unlabeled data
- RL: maximization of a reward signal instead of trying to find hidden structure
  - Finding such a structure in RL does help its learning, but it does not solve the RL problem itself



Unsupervised learning

Difference between RL and other ML

- There is no supervisor, only a reward signal
- Feedback is delayed, not instantaneous
- Time really matters (sequential, non I.I.D data)
- Agent's actions affect the subsequent data it receives

IID: Independent and Identically Distributed